PR 2 Desirable Judgement:

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☆ Theoretical foundation

① What is inferential statistics?

: Inferential statistics is the branch of statistics that allows us to make predictions or draw conclusions about a larger group by analyzing a smaller sample of data taken from it.

② What is hypothesis testing and it's Components?

: Hypothesis Testing is a statistical method used to make decisions about population parameters based on sample data.

: It helps you determine whether an assumption about a population is true or not

- Components of Hypothesis Testing.

① Null Hypothesis (H_0)
② Alternate Hypothesis (H_1 or H_a)

☆ Null Hypothesis (H_0)

: It is the default or initial claim.
: It assumes no effect, no difference or status quo.

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Alternate Hypothesis (H₁ or Ha):

: It is what you want to prove or test
: It represents a change, an effect, or a difference.

③ Explain Confidence interval and critical value.

: Confidence interval.

: A Confidence interval gives a range of values that is likely to contain the true population parameters based on sample data.

: Critical value.

: A critical value is the cutoff point on the test statistic's distribution that defines the boundary between accepting or rejecting the null hypothesis.

④ P-value

: The P-value is the probability of obtaining result at least as extreme as the observed result under the assumption that the null hypothesis is true.

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⑤ Type I and Type II errors.

: A Type I error occurs when the null hypothesis is rejected when it is actually true with probability α.

: A Type II error occurs when the null hypothesis fails to be rejected when it is actually false, with probability β.

⑥ Z-test, t-test, chi-square test, and ANOVA test.

- Z-test
: Compares sample means when the population variance is known and the sample size is large ($n \geq 30$).

- t-test
: Compares means when the population variance is unknown and the sample size is small ($n \geq 30$)

- chi-square (χ^2) test :
: Analyzes categorical data to check for relationships between two variables or to see if observed counts match expected distributions.

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- ANOVA

: Tests for statistically significant differences among the means of three or more independent groups simultaneously to avoid the accumulated error of multiple separate t-test.

⑧ Covariance.

: Measures the direction of the relationship between two variables - whether they move in the same direction or opposite direction without indicating how strong the link is.

⑨ Correlation

: Measures both the direction and strength of a linear relationship between two variables on a standard scale from -1 to +1